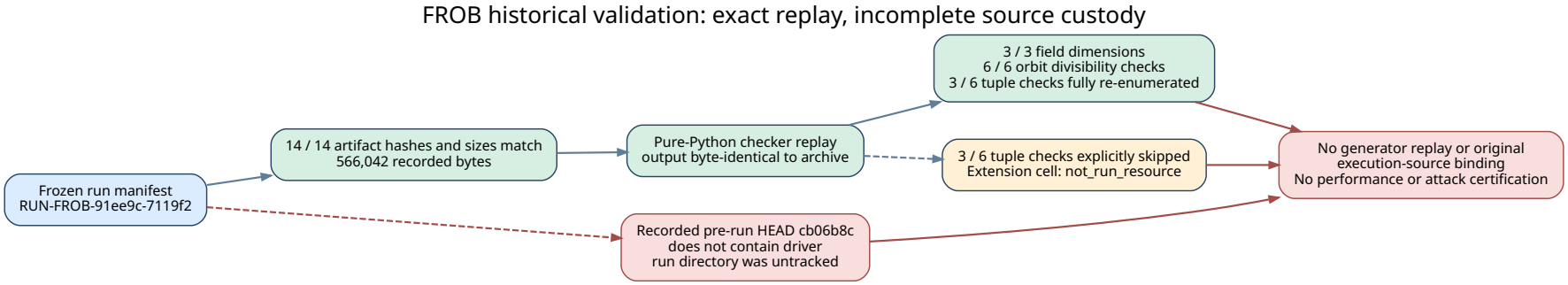
# Historical FROB checker replay and source custody

Validation audit, 22 October 2026

**Scope.** This checks frozen run `RUN-FROB-91ee9c-7119f2` at source tree `e83bb2f14414c9669360816aeee5876dfcfd4384`. It is an integrity audit, not a new FROB measurement or a research verdict. The run directory was copied before executing its pure-Python checker; frozen bytes were not edited.

| Check | Result | Limit |
|---|---|---|
| Archived artifact hashes and sizes | 14/14 match; 566,042 bytes | Storage integrity only |
| Checker output | Byte-identical replay | Same archived checker source |
| Field dimensions; orbit divisibility | 3/3; 6/6 | Recorded core cells |
| Tuple point/order/Frobenius checks | 3/6 confirmed | Other three explicitly skipped |
| Extension cell | `not_run_resource` | 540-second historical checkpoint |




**Source custody.** The manifest records pre-run HEAD `cb06b8c20c2d6559041c45c4a4b29dd7f132fdf5` and says the run directory was untracked at execution start. The driver is absent from that commit. Current-schema `code.commit`, `code.command`, and `environment` fields are absent, although separate command and environment artifacts are hash-pinned. Their presence does not bind the executed driver to the recorded HEAD.

**Boundary and next check.** The replay does not rerun the Sage generator, the three skipped tuple enumerations, or any performance experiment. Recover a contemporaneous execution snapshot and exact command if available; otherwise run a fresh source-bound replay against the frozen specification and controls. The Coordinator must adjudicate the original record without assigning the pre-run HEAD as a false execution-source commit. Machine receipt: `docs/reports/2026-10-22-frob-provenance-audit/audit.json`.